

Krea 2 → Scene-Linear HDR true HDR assets from a small LoRA

Making Krea 2 (12.9B, Qwen Image VAE) generate true scene-linear HDR (EXR, values $\gg 1$) via a small LoRA, validated end-to-end into ComfyUI. This report records what was built and measured, with real curves, tables, and image analysis.

Five core conclusions (all measured):

- **The route works** — "LogC4 encoding + frozen VAE + a DiT LoRA on RAW": Krea 2 learns to generate extended dynamic range.
- **Learned, not fabricated** — same prompt+seed, the base clips highlights flat (fake), the LoRA spreads them (real): pinned-at-ceiling 6.20% → 0.66%.
- **Generalizes** — 7/8 out-of-distribution prompts (fire/neon/portrait/cathedral/candle) produce real HDR with the bright region on the light source.
- **Runs in ComfyUI** — LoRA needs a key remap (0→264) + two nodes; e2e yields a real HDR EXR (459× / 8.7 stops).
- **Honest ceiling** — single-plane LogC4 caps peak radiance $\sim 470\times$; sun magnitude needs a multi-plane scheme.

1. Method & pipeline

The VAE is untouched (keeping the official RAW→Turbo transfer). Scene-linear radiance is compressed into $[0,1]$ with **ARRI LogC4** and fed to the frozen Qwen Image VAE; the model learns to generate in LogC4 space; **inverse-LogC4** recovers linear light for EXR. VAEDecode's $[0,1]$ clamp is harmless (the output is a $[0,1]$ LogC4 code; >1 appears only after inverse-LogC4).

train: EXR → (median→0.18) → LogC4 $[0,1]$ → 8-bit PNG → VAE latent → DiT LoRA(RAW)

infer: prompt → Krea2+LoRA → LogC4 $[0,1]$ → inverse-LogC4 → scene-linear($\gg 1$) → EXR / HDR AVIF

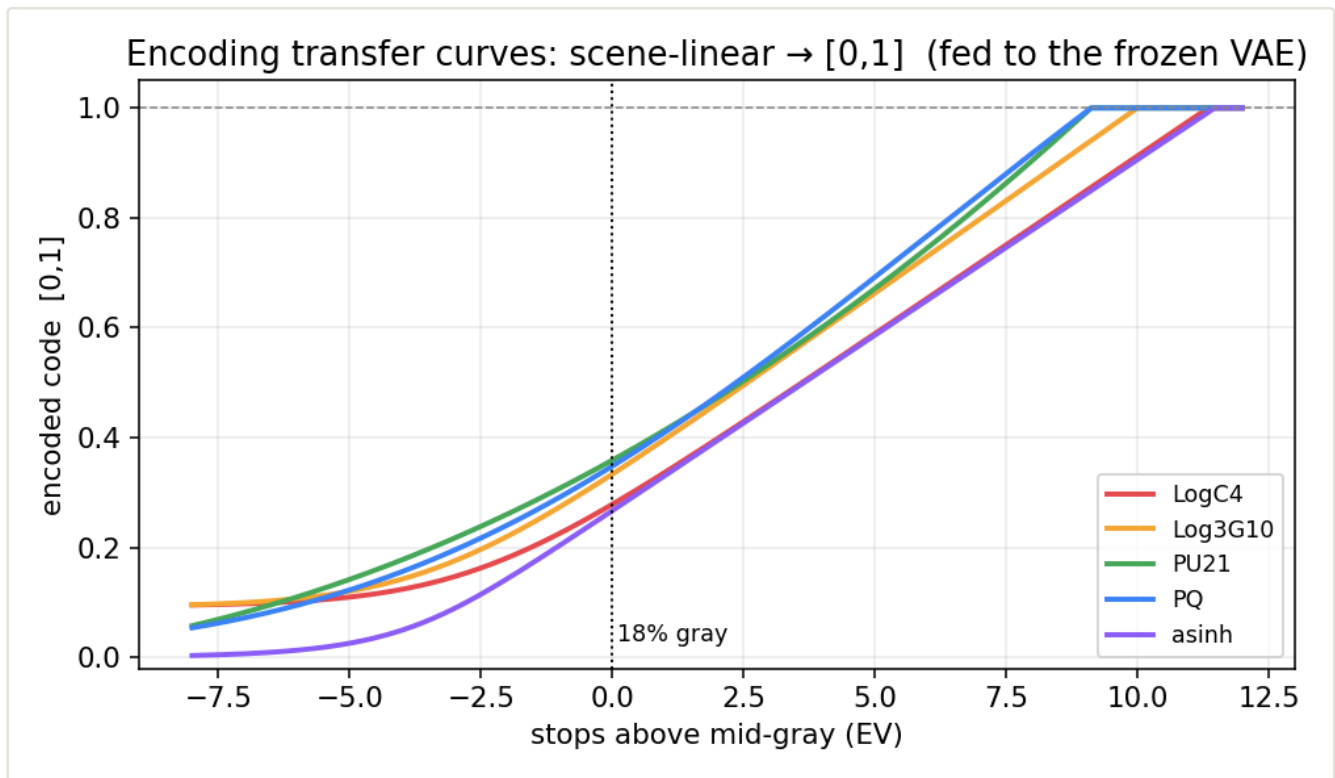


Fig 1. Five encoding curves mapping scene-linear to $[0,1]$. Camera logs (LogC4/Log3G10) place 18% gray near 0.28-0.33, close to sRGB 0.46 (on-manifold); each curve has a different ceiling (linear peak at code 1).

2. CPU validation (no GPU)

Before any training code, the linear $\leftrightarrow$ encoding mapping and the EXR chain were validated numerically. **All 8 curves invert to $\sim 1e-13$ (float64).** Mid-gray placement: LogC4 0.278, Log3G10 0.333, PU21 0.359, PQ 0.348 (near sRGB 0.46); mu-law 0.015 and Log-Gamma 0.809 are badly placed — confirmed poor.

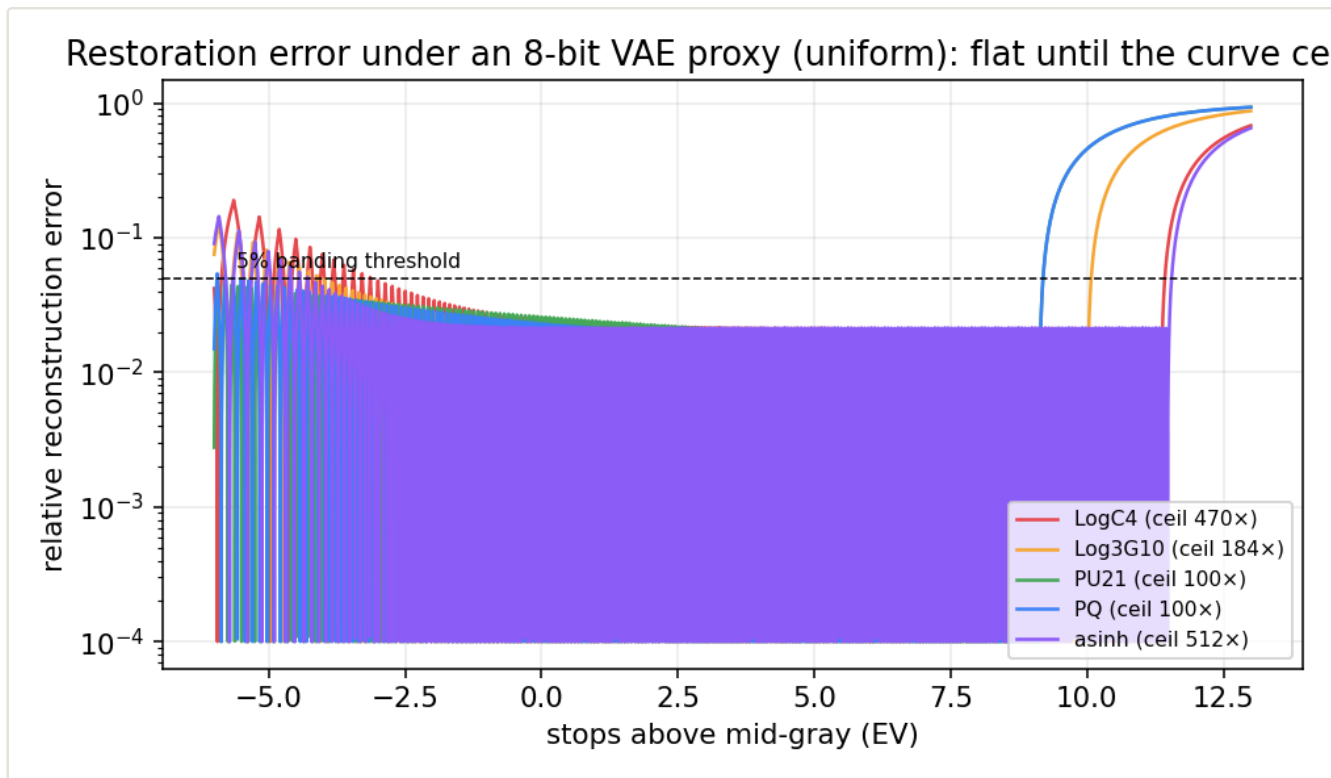


Fig 2. Reconstruction error vs exposure under an 8-bit VAE proxy. Log curves stay ~2% flat until their ceiling (~17 codes/stop, no banding wall) — correcting the earlier "85 codes -> 4-6 stops" oversimplification. The real limiter is the VAE's non-uniform precision + the sun overflowing the ceiling.

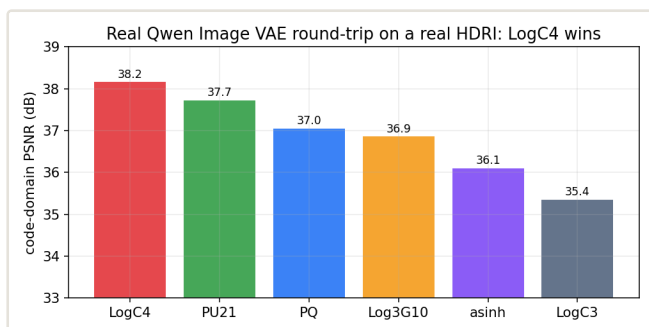


Fig 3. Real Qwen VAE round-trip code-PSNR on a real HDRI: LogC4 (38.2 dB) best, best bright band. X2HDR premise confirmed: perceptual encoding ~2% band error, while naive scaled-linear fed to the VAE gives **7211%** highlight error.

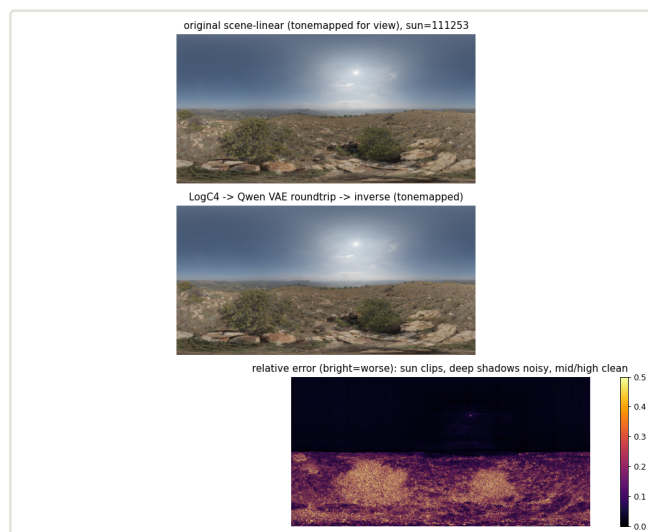


Fig 4. LogC4 -> real VAE -> inverse on a real HDRI: tonemapped original and reconstruction are near-identical (top/mid); error concentrates in dark textured foreground (~22%); the sun clips. Clean window is mid+high ~2%.

First-hand: the Qwen VAE decoder has **no tanh/sigmoid** — its only bound is `torch.clamp(-1,1)`, so the bottleneck is the encoder, not the decoder. Real data: Poly Haven klooferdahl max 111,253×, 31.6 stops, median 0.178; S2R-HDR sample peaks at only 21× with no sun (a hygiene source, not an IBL/sun backbone). EXR chain: 32-bit ZIP preserves 4000× exactly; half overflows above 65504 -> +Inf; clamp->[0,1] and implicit sRGB destroy HDR (4000 -> 1.0 / 33.4).

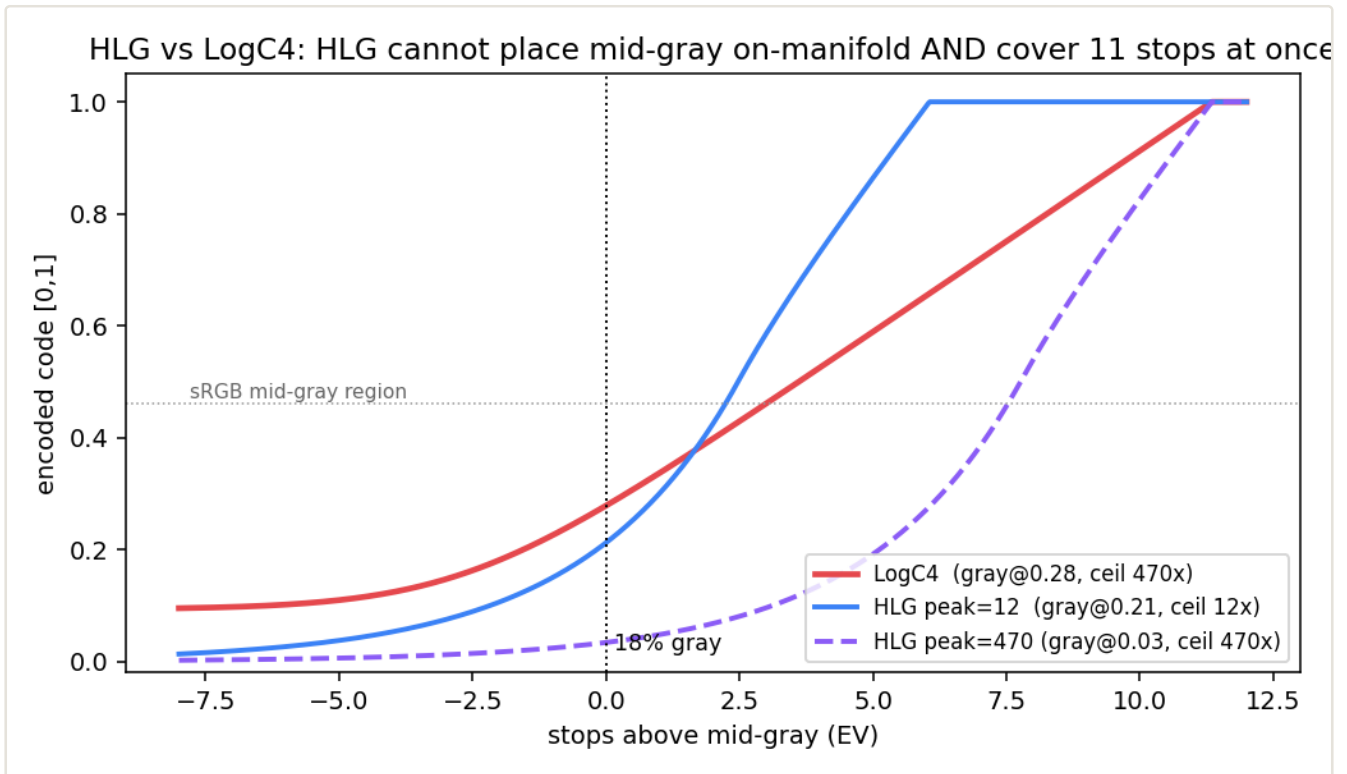


Fig 5. HLG vs LogC4. HLG cannot place mid-gray on-manifold AND cover 11 stops at once: at a usable mid-gray its ceiling is only 12x (bright highlights clip); at LogC4's 470x ceiling it crushes mid-gray to ~0.03 (off-manifold). LogC4 uniquely does both.

3. Dataset

40 CC0 Poly Haven HDRIs re-projected into **normal perspective scene images** (not panoramas); 320 crops, 229 HDR-rich; LogC4-encoded 8-bit PNG + caption + manifest (reproducible float crops).



Fig 6. Per-image HDR-content check: same PNG read naively as SDR -> peak 1.0 (flat/grey); read correctly via inverse-LogC4 -> peak 470x (sun and gradation present). 93% of training images decode to real HDR.

A real dataset flaw: 50% of "with visible sun" captions sit on crops with no sun in frame; parking-lot content and the sun ended up in separate crops, so the model can't compose "parking lot + sun". Fix: honest per-crop captioning (only write "sun" when the crop's measured peak is high) + wider FOV so sun and scene co-occur.

4. Training & pipeline audit

musubi-tuner krea2_train_network.py: RAW + lora_krea2 dim/alpha 32 (264 modules), lr 1e-4, adamw8bit, 10 epochs / 2290 steps. **Training VRAM ~34 GB** (bf16 DiT + adamw8bit + gradient checkpointing + batch 2 @ 1024²); fits 24GB with fp8 + block swap.

Framework	Image->VAE normalization	Color aug	HDR-safe?
musubi	/127.5-1 (validated [0,1]->[-1,1])	none	safe (recommended)
ai-toolkit	image*2-1 correct	ships ColorJitter	disable color aug (else corrupts LogC4)

Rule: never apply a color op to LogC4 pixels. Under this route the trainer needs no HDR-awareness — the HDR lives in the encoding.

5. Results

5.1 Real vs fabricated (core evidence)

">1 alone isn't proof — the decode curve inflates any image. The test is highlight clipping: the base clips highlights flat at the ceiling (fake HDR); the LoRA spreads them into graduated values (real HDR).

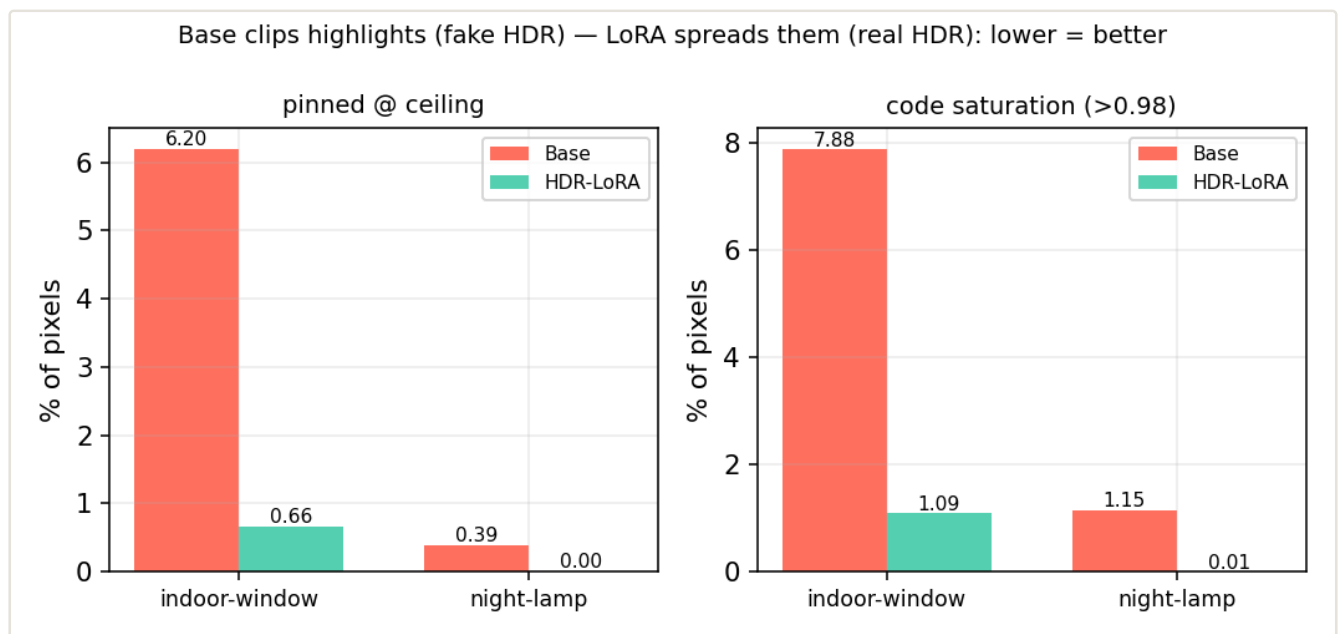


Fig 7. Same prompt+seed, base vs LoRA. Both metrics (pinned-at-ceiling / code saturation) are far lower for the LoRA -> highlights keep real gradation.

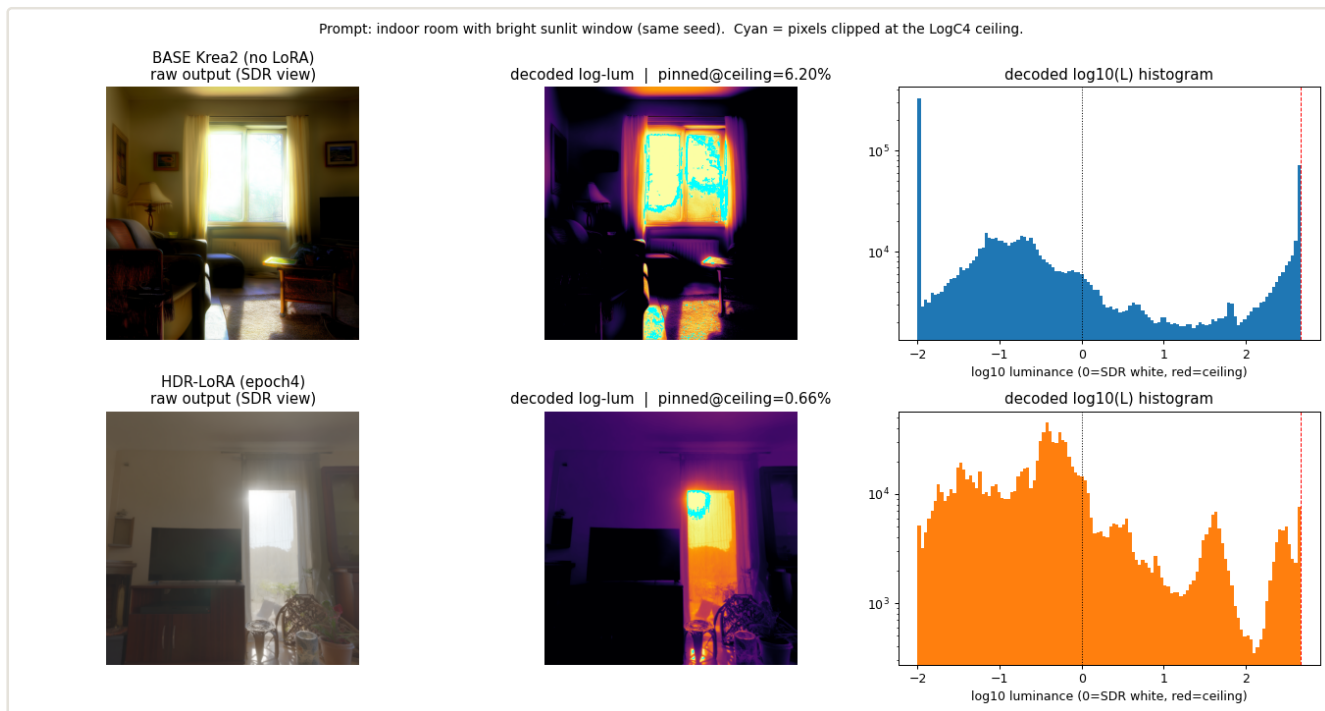


Fig 8. Window close-up: the base blows the window to a flat ceiling slab (6.20% pinned, a histogram spike at the ceiling); the LoRA holds graduation (0.66%, a smooth highlight distribution). Direct evidence the backbone learned extra information.

5.2 Generalization (epoch 10, out of distribution)

With only the HDR trigger phrase, on community prompts (portrait/fireplace/neon/dragon/cathedral/candle). **7/8 produce real HDR** ($\leq 0.01\%$ pinned, light on the source). The candle that failed at epoch 8 is fixed at 10 (flame 400x). The lone exception (spaceship) is not a black frame — a dim deep-space starfield with no bright source (content, but no extended range).

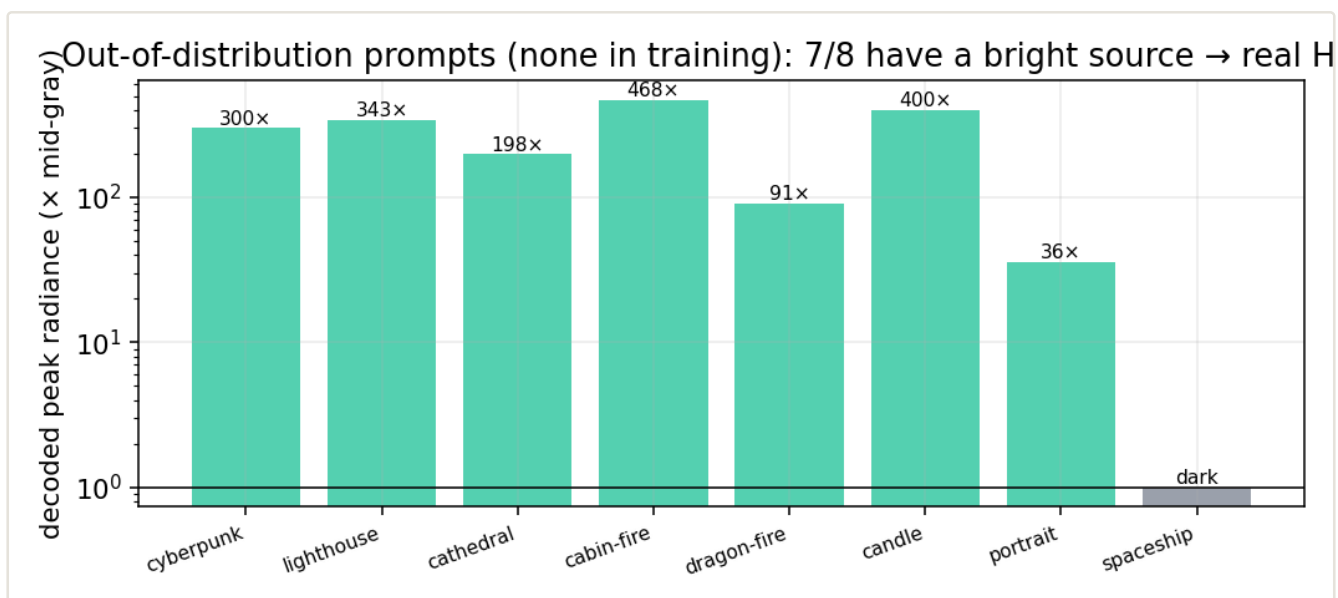


Fig 9. Decoded peak radiance per OOD scene (log axis). Green = has a bright source → real HDR (7); grey = dark scene, no source (spaceship starfield).

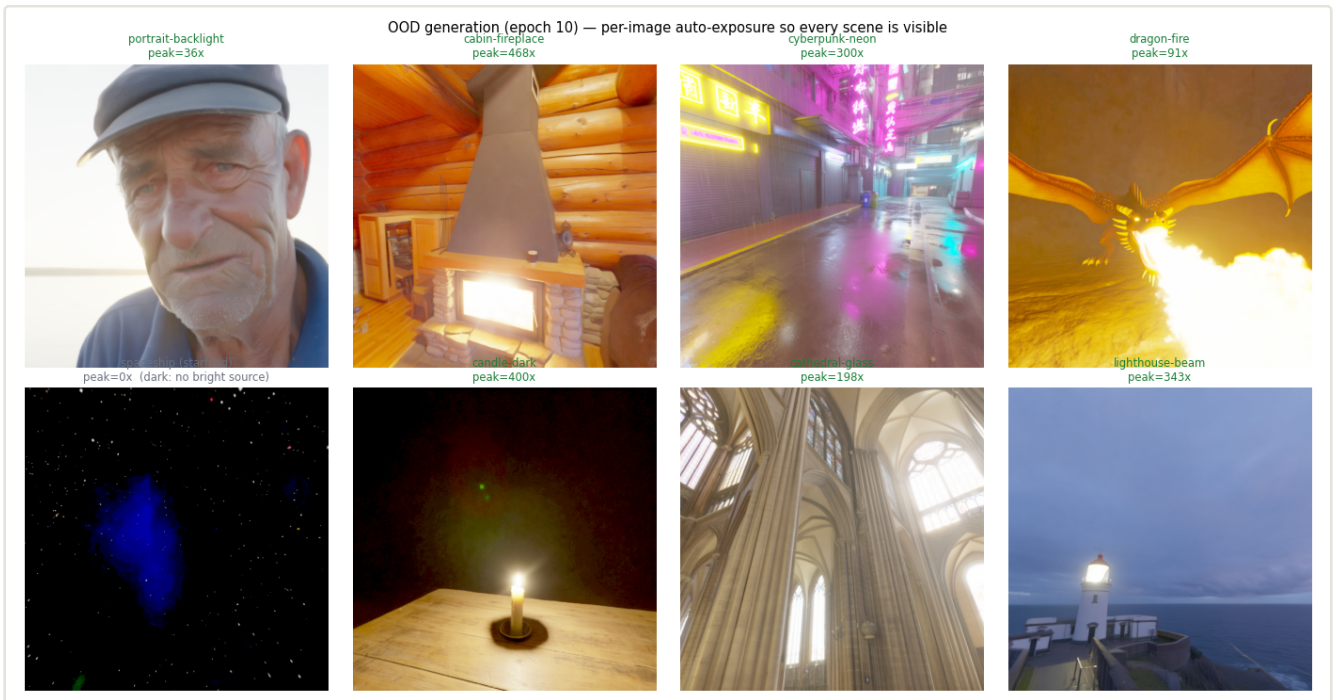


Fig 10. OOD generation (per-image auto-exposure so every scene is visible): fire/neon/stained-glass/lighthouse highlights land on the correct source; the candle is a lit candle (400x); the spaceship is a dim starfield.

5.3 End-to-end in ComfyUI



Fig 11. Full ComfyUI graph output: sunset over ocean, sun at the horizon as the bright region, peak 459x / 8.7 stops — quality on par with musubi.

5.4 Color check & display fix

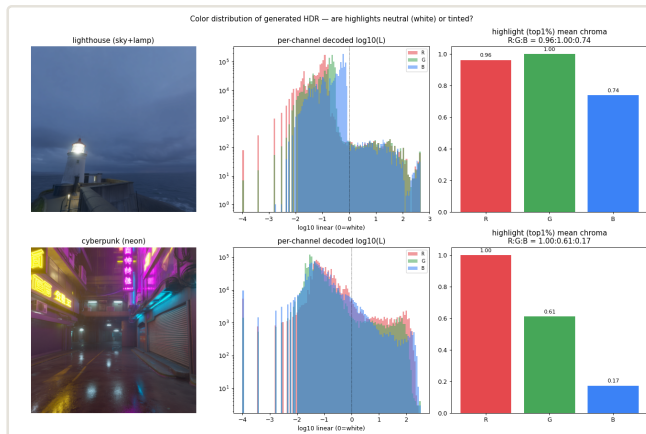


Fig 12. Highlights are neutral, not blue-tinted (lighthouse lamp R:G:B = 0.96:1.00:0.74).

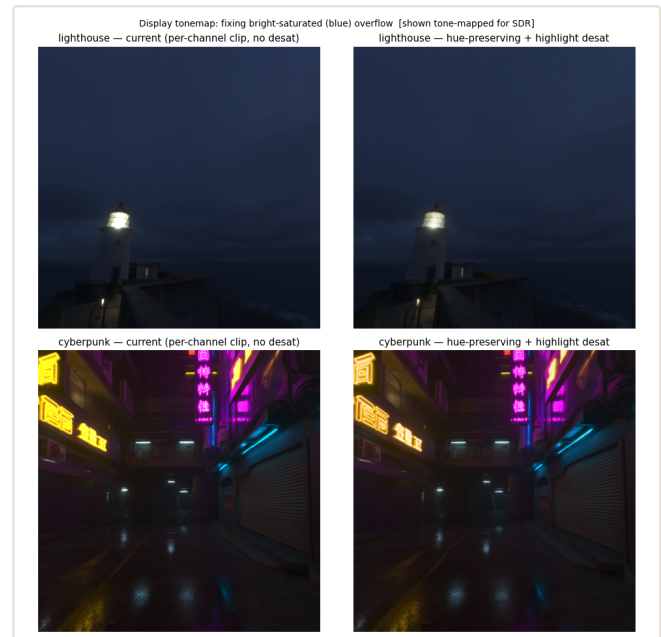


Fig 13. The "blue overflow" was display-side: per-channel hard clip (left) -> hue-preserving roll-off + highlight desaturation (right). The EXR is unaffected.

6. ComfyUI landing (end-to-end verified)

Question	Answer
musubi LoRA loads directly?	No — ComfyUI's own loader matches 0/264 keys (naming differs)
After remap?	Yes — remap_lora_for_comfyui.py maps 264/264; ComfyUI attaches 263 patches
Built-in SaverImage for EXR?	No — 8-bit + clamp, HDR lost
Fix	Two nodes: LogC4->Linear + Save EXR Scene-Linear (also a LogC4 option contributed to built-in SaverImageAdvanced)
End-to-end	Yes — real EXR peak 459x / 8.7 stops
Gotcha	Do not add ModelSamplingSD3 — Krea2 already shifts; a second one washes the image out

7. Conclusion, limits, next steps

Conclusion: the first-step route holds; the full chain is proven and lands in ComfyUI.

Limits: single-plane ~470x cap; quality limited by the small dataset (40 HDRIs); caption/crop flaws; RAW->Turbo transfer untested.

Next (by payoff): (1) data — honest captioning + wider FOV so sun and scene co-occur; scale to 978 Poly Haven + Fairchild + procedural. (2) sun magnitude — multi-plane / gain map or exposure stack. (3) sky — a parametric sun+sky model. (4) attack fabrication at the representation level — train the VAE encoder (16->32), at the cost of the free Turbo transfer.

Appendix: key numbers

Item	Value
Encoding round-trip error	~1e-13 (float64, 8 curves)
Real-VAE code-PSNR	LogC4 38.2 dB (best); naive linear highlight error 7211%
Real HDRI	kloofendal 111,253x / 31.6 stops / median 0.178; S2R-HDR 21x (no sun)
Dataset	320 imgs / 229 HDR-rich; 93% decode to real HDR; 50% "sun" mislabeled
Training	264 LoRA modules, dim/alpha 32, 2290 steps, 10 epochs, ~34 GB VRAM
Real-vs-fabricated (window)	pinned-at-ceiling 6.20% -> 0.66%; saturation 7.88% -> 1.09%
Generalization	7/8 OOD produce real HDR
ComfyUI	LoRA match 0->264 (after remap), 263 patches; e2e EXR 459x / 8.7 stops
Single-plane ceiling	~470x (LogC4)

Companion: research-findings.md (route survey), tools/ (validation suite), comfyui/ (nodes + remap + workflow). LoRA and dataset are public on Hugging Face. 2026-07.